

GEOGRAPHY

PHYSICAL GEOGRAPHY



CLIMATOLOGY

MODULE – 38

WORLD CLIMATIC TYPES

THE STEPPE CLIMATE

- **VEGETATION AND ECONOMY OF PEOPLE**

5

THE TEMPERATE CONTINENTAL (STEPPE) CLIMATE

NATURAL VEGETATION



TEMPERATE STEPPE GRASSLANDS

1 Our reference to Steppe grassland is taken to mean the temperate grasslands of the mid-latitudes, the Steppes, Prairies Pampas, Veld and Downs.

2 It is natural to expect the Steppe to be grasslands covered, differing only in the density and quality of the grass.

3 Their greatest difference from the tropical savanna is that they are practically treeless and the grasses are much shorter.

5

THE TEMPERATE CONTINENTAL (STEPPE) CLIMATE

NATURAL VEGETATION



CHERNOZEM SOILS OF STEPPE

4

Where the rainfall is moderate, above 50 cms, the grasses are tall, fresh and nutritious and are better described as long Prairie grass.

5

This is typical of the grass of the Wheat-lands in North America, the rich black earth or chernozem areas of Russian Ukraine and the better watered areas of the Asiatic Steppes.

6

Where the rainfall is light (less than 20 inches) or unreliable, or the soil is poor, as in the continental interiors of Asia the short steppe type of grass prevails.

5

THE TEMPERATE CONTINENTAL (STEPPE) CLIMATE

NATURAL VEGETATION



TROPICAL SAVANNAH

7

The grasses are not only shorter but also **wiry and sparse**, often found in discontinuous clumps, **with bare soil exposed between them**.

8

These areas are less suitable for arable farming and are used **for some form of ranching as in the High Plains of USA**.

9

The climatic requirements of grass are quite different from those of trees.

5

**THE TEMPERATE CONTINENTAL
(STEPPE) CLIMATE****NATURAL VEGETATION**

10 They require less moisture than trees and an annual precipitation of 10 to 20 inches is adequate.

11 Their growth is not abruptly checked by summer droughts or winter cold.

12 The steppe grass can lie dormant throughout the prolonged drought.

13 They sprout and come to life as soon as the temperature is warm enough for plant growth and grow steadily with very little moisture.

14 The appearance of the temperate grasslands varies with seasons.

15 In spring, the grass begins to appear, green, fresh and blooming with colorful flowers.

16 The light rainfall that comes in late spring and early summer greatly stimulates their growth and there is plenty for the animals to graze. The herdsmen are busiest at this time.

5

THE TEMPERATE CONTINENTAL (STEPPE) CLIMATE

NATURAL VEGETATION

17 In summer, there is so much heat and evaporation that the grass is scorched.

18 The carpet of bluish-green grass turns yellow and soon brown.

19 Towards autumn, the grass withers and dies, but the roots remain alive and lie dormant throughout the cold winter.

20 The winter is harsh and long, but the snow is never of great depth.

21 Everything is quiet but with the next spring, the cycle is repeated and the steppe is alive again.

5

THE TEMPERATE CONTINENTAL (STEPPE) CLIMATE

NATURAL VEGETATION



PRAIRE GRASSLANDS OF USA

22 In the cultivated regions, such as the wheat farms of Prairie, double rows of trees are planted around the house to shield the occupants from the strong winds which come unobstructed across miles of level ground.

23 This provides the greatest contrast in a land which is essentially grass,

24 There are no hedges and few fences and the rows of planted trees form an unusual landmark from the air!

5

**THE TEMPERATE CONTINENTAL
(STEPPE) CLIMATE****NATURAL VEGETATION**

25 Trees are very scarce in the steppes, because of the scanty rainfall, long droughts and severe winters.

26 The rolling plain is an endless stretch of grass, whether green or brown, except along the water courses where a few low willows, poplars or alders break the monotony.

27 Polewards an increase in precipitation gives rise to a transitional zone of wooded steppes where some conifers gradually appear.

28 Even then, the trees are very scattered and few in number. Towards the equator, the steppe grass becomes shorter and sparser, till it merges into the desert with thorny scrub.

5

THE TEMPERATE CONTINENTAL (STEPPE) CLIMATE

ECONOMIC DEVELOPMENT



BISON OF PRAIRIES

1 The temperate grasslands were once the home of grazing animals: **wild horses in the Asiatic Steppes**, **swift-footed bison in the Prairies** and **untamed buffaloes in the Pampas**.

2 Even as recently, as the last century, these grasslands were dominated by **nomadic and semi-nomadic peoples like Kirghiz of Asiatic steppes**.

3 Cultivation was unknown and the region was one of the most sparsely populated parts of the world.

5

THE TEMPERATE CONTINENTAL (STEPPE) CLIMATE

ECONOMIC DEVELOPMENT



ROLLING PLAINS OF PRAIRIES

4

In recent years, great changes have taken place in the grasslands and few areas, in fact, have managed to retain their original landscape.

5

The grasslands have been ploughed up for **extensive, mechanized wheat cultivation** and are now the “granaries of the world”.

6

Besides wheat, **maize** is **increasingly cultivated** in the warmer and wetter areas.

5

THE TEMPERATE CONTINENTAL (STEPPE) CLIMATE

ECONOMIC DEVELOPMENT



ALFALFA GRASS

7

The **tufted grasses** have been replaced by **nutritious lucerne or alfalfa grass** for cattle and sheep rearing.

8

These temperate grasslands are now the **leading ranching regions of the globe**. We shall now describe more closely each of these economic activities

5

THE TEMPERATE CONTINENTAL (STEPPE) CLIMATE

ECONOMIC DEVELOPMENT

Various types of economic activities found in the Mediterranean regions.

TYPE- 1

NOMADIC HERDING



1 This type of migratory animal grazing **has almost disappeared** from the major grasslands.

2 The herders were wandering tribes e.g. the **Kirghiz, Kazakas and the Kalmuk.**

3 They used to travel over long distances like the Bedouin of the Arabian Deserts, in search of grass and water for their animals-cattle, sheep, goats and horses.

5

THE TEMPERATE CONTINENTAL (STEPPE) CLIMATE

ECONOMIC DEVELOPMENT

NOMADIC HERDING

TYPE- 1

4 From these domesticated animals they obtained meat, milk, wool, hides, bines, and horn.

5 You would be surprised at the number of things they made out of these.

6 The **wool** was woven into felt for **tents and garments**.

7 The **leather** was used for making **boots, saddles, and belts** which were very essential in a country where the chief riding animal was the horse.

8 The **bones and horns** were not wasted but made into **tools, utensils and weapons**.

9 Many of their home-made products were exchanged at trading posts or with the caravans for guns, canned food, grains, tea, coffee, sugar, medicines and other essential goods.

WE EXPRESS OUR
SINCERE THANKS
FOR VIEWING THIS VIDEO

Presented by



Learning Space

For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

OUR TEAM

G. V. Rao

K. Srikanth

D. Sunil Kumar

L. V. Krishna

Y. Deepthi

Amrita Naidu

M. Binukrishna

G. Ravi Babu

D. Joji

K. Victor Babu

Visit us at:

www.learningspacedigital.com

0 8 6 6 -2 4 4 4 4 7 2

0 9 8 4 9 9 4 2 2 9 9